T TALENTTRACE AI

TalentTrace AI
Evidence-led recruitment

From job-specific CV screening to reviewed interviews and traceable assessment

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Project overview

One connected workflow across three stages of hiring.

01

PRE-INTERVIEW

Define a role, ingest CVs, extract information, score and rank applicants.

02

DURING INTERVIEW

Conduct a secure browser interview and capture structured dialogue.

03

POST-INTERVIEW

Analyse Q&A evidence and produce a recruiter-reviewable assessment.

POSITIONING **The AI supports the recruiter; the recruiter retains the final judgment.**

Problem statement

Hiring volume and fragmented evidence make consistent assessment difficult.

High CV volume

Reviewing every application manually is slow and difficult to scale.

Inconsistent judgments

Different reviewers may apply different standards to similar candidates.

Unstructured interviews

Interview evidence can remain in scattered notes instead of searchable records.

Opaque decisions

Scores and recommendations are hard to defend without traceable sources.

THE DESIGN RESPONSE

Structured inputs

→

Explainable analysis

→

Human review

Project objectives

Intended outcomes. Acceptance results and remaining gaps appear later in this deck.

01	**Reduce manual screening effort**	Screen and rank a CV batch against a job profile.
02	**Increase consistency**	Use saved, reviewable criteria and a scoring breakdown.
03	**Structure interview content**	Capture speaker-labelled, timestamped turns.
04	**Make assessments defensible**	Link conclusions to CV pages and interview segments.
05	**Support faster decisions**	Bring pre- and post-interview evidence into one report.

Key functionalities

Implemented capabilities with clearly identified acceptance boundaries.

F1 CV screening & ranking

Groq-assisted extraction, OCR, job-specific comparison.

Real synthetic tests

F2 Weighted candidate scoring

Criteria, saved scores, breakdown and evidence.

Real synthetic tests

F3 Live video interview

Recruiter/candidate access, scoped LiveKit media.

2-party Cloud verified

F4 Speech transcription

Role-labelled turns, consent and optional storage.

Live browser STT

F5 Q&A evaluation

Answer quality, question pairing, requirement coverage.

Synthetic model-double

F6 Consolidated report

CV evidence plus authorized interview findings.

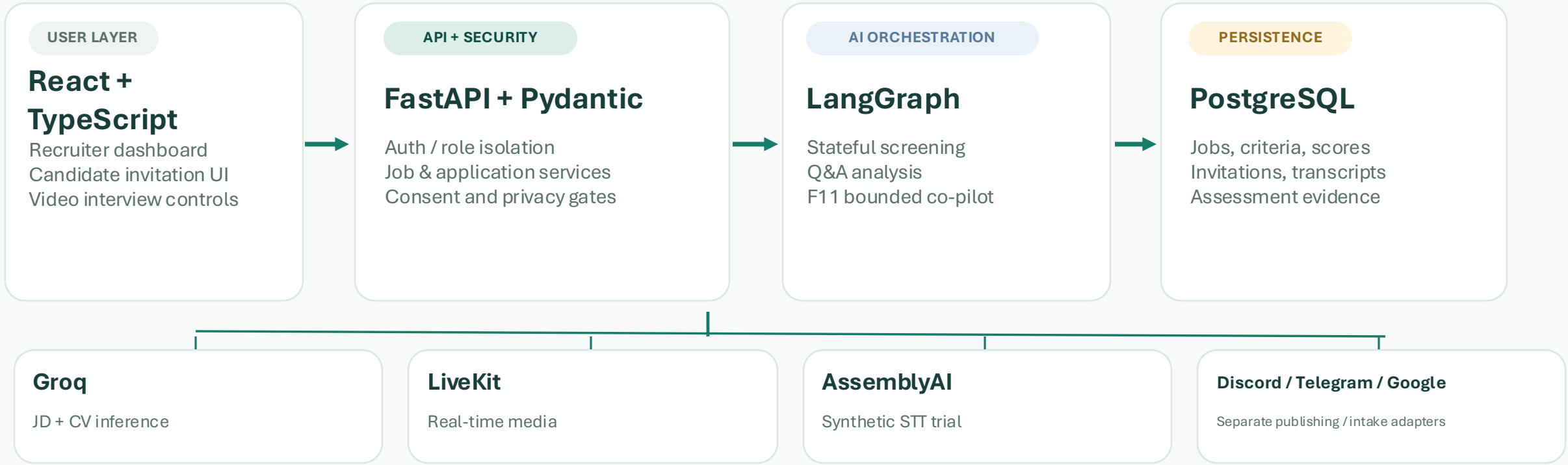
Synthetic report / print verified

F11 | DYNAMIC INTERVIEWER CO-PILOT

Recruiter-only LangGraph suggestions; fake-STT browser lifecycle PASS. Real AssemblyAI chain not yet verified.

System architecture

Application components and explicit data-processing boundaries.



Third-party calls only when configured and authorized • synthetic data used in evaluation

Technology stack

Tools were selected for explicit orchestration, testability and local reproducibility.

FRONTEND — **React, TypeScript, Vite**

Browser UI, responsive workflow, typed API views

BACKEND — **Python, FastAPI, Pydantic**

APIs, contracts, access control and validation

AGENTIC AI — **LangGraph, Groq API**

Explicit state transitions + approved free-tier inference

DATABASE — **PostgreSQL + pgvector**

Persistent jobs, CV evidence, scoring and transcripts

LIVE INTERVIEW — **LiveKit / WebRTC**

Scoped real-time audio and video transport

TRANSCRIPTION — **AssemblyAI; existing local Whisper path**

Speaker-labelled speech pipeline; live completion pending

DOCUMENTS — **PyMuPDF + Tesseract**

PDF text extraction and scanned-page OCR

DELIVERY — **Docker Compose, Git, GitHub**

Versioned services, reproducible setup, traceable commits

Workflow / process flow

Connected recruiter journey; the full live F4-to-F5/F11 path remains partial.

1

Create and review job

Groq JD → stable Job Reference

2

Add candidate CVs

Upload / approved imports; OCR

3

Screen and review

Groq analysis + weighted evidence

4

Select and book

Human review → scheduled slot

5

Conduct interview

LiveKit recruiter + candidate

6

Interpret interview

Consented STT → grounded Q&A

7

Review assessment

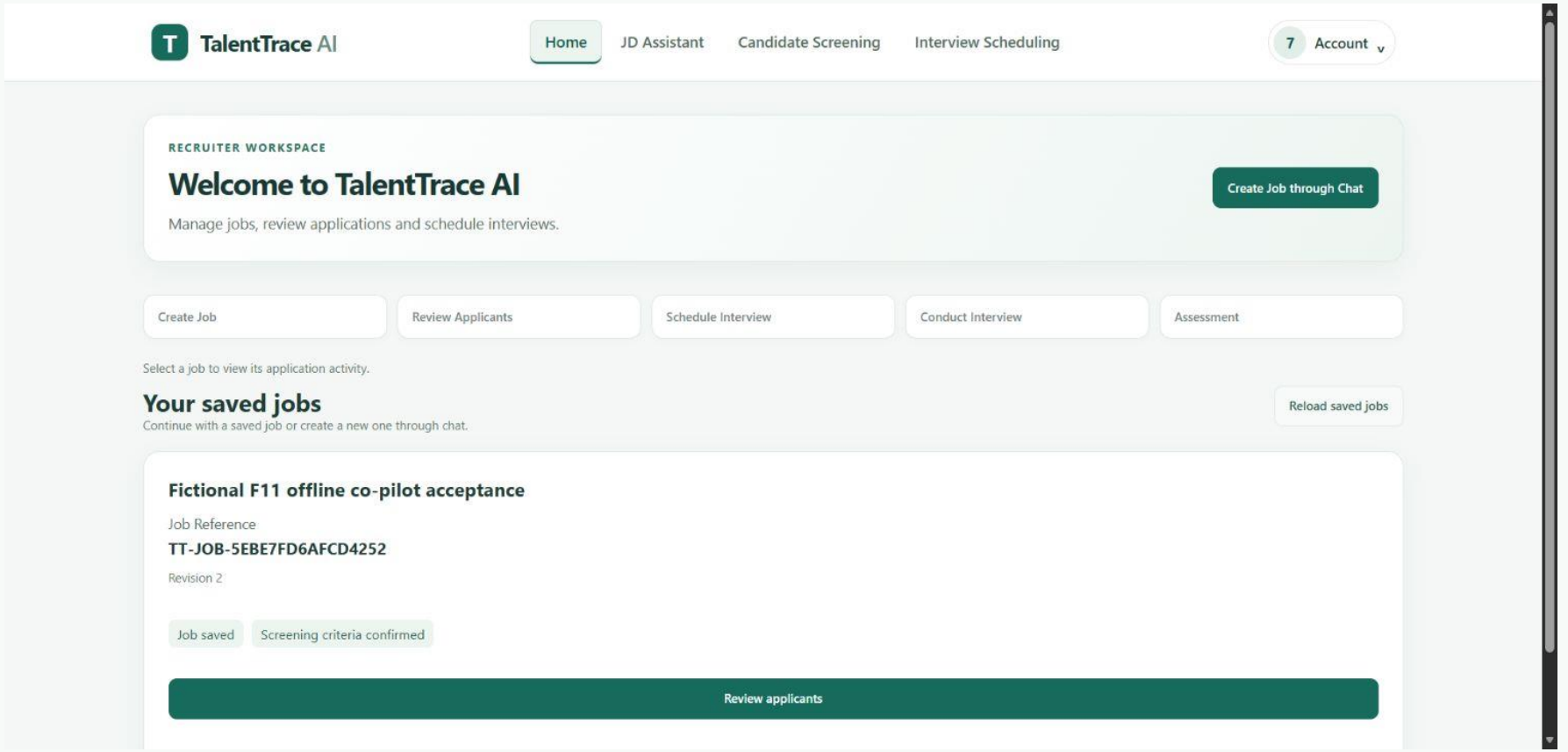
CV + interview evidence → report

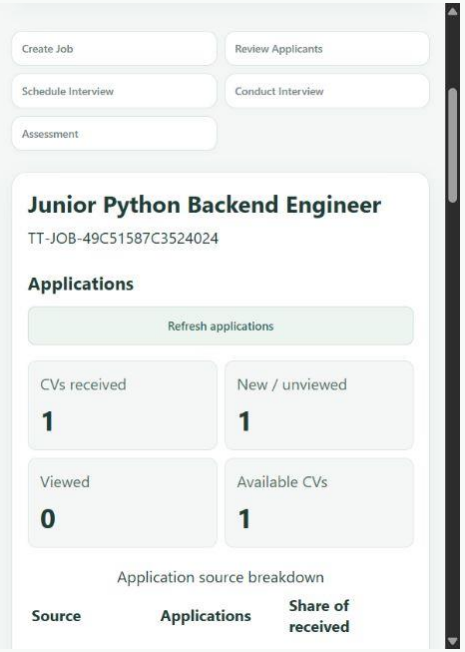
F11 offers optional provisional follow-ups. Recruiters review evidence and make the hiring decision.

UI demonstration: Home and applications

Actual local browser captures. The desktop and mobile examples use separate synthetic fixtures.

Saved jobs and a clear next action



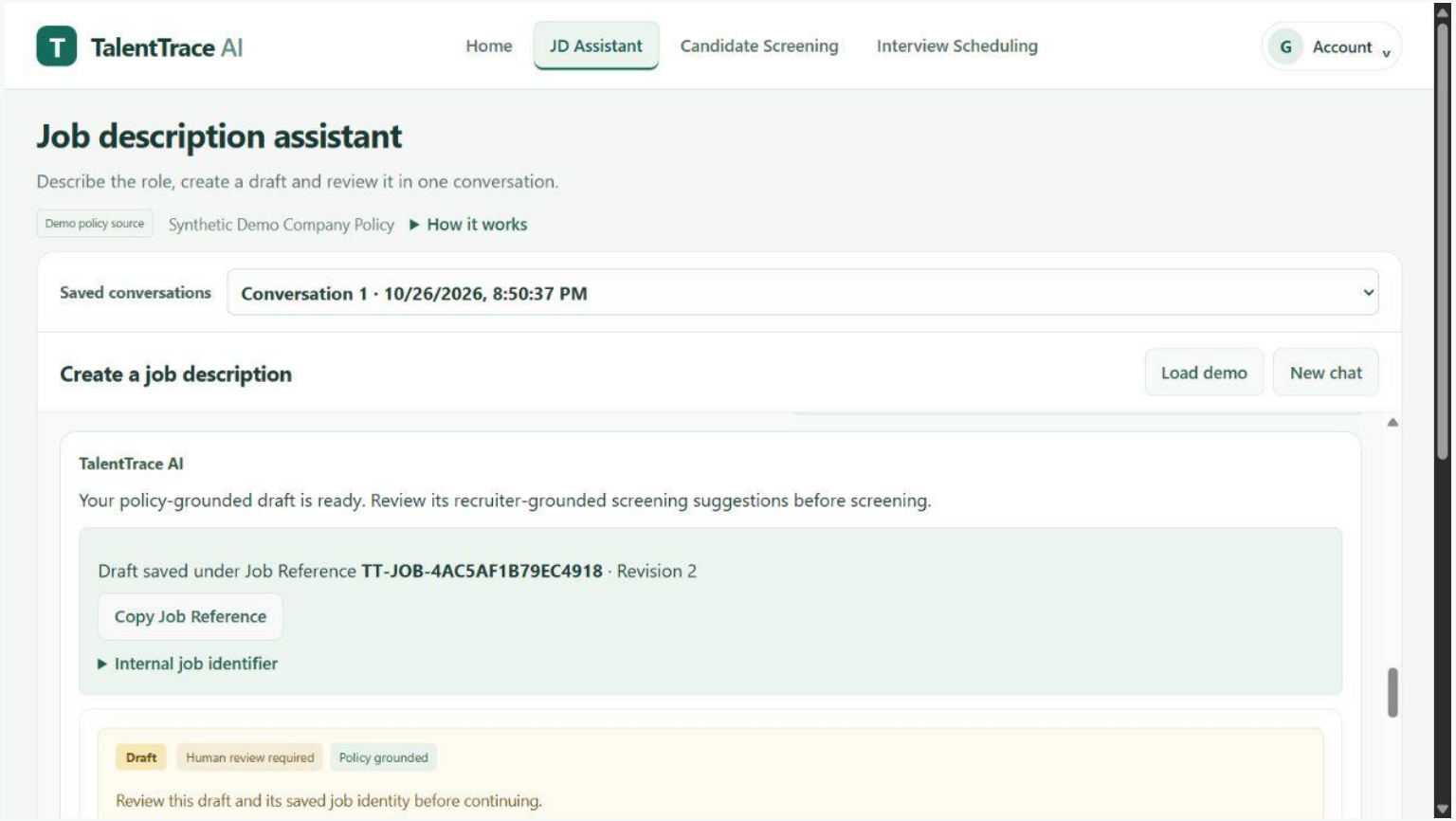


One approved Google import

1 received / 1 new / 1 available
Snapshot at import, not automatic sync

UI demonstration: saved JD refinement

Actual Groq-demo browser capture after refresh. The saved job remains at revision 2.



What to explain

A conversation refines the existing saved job.

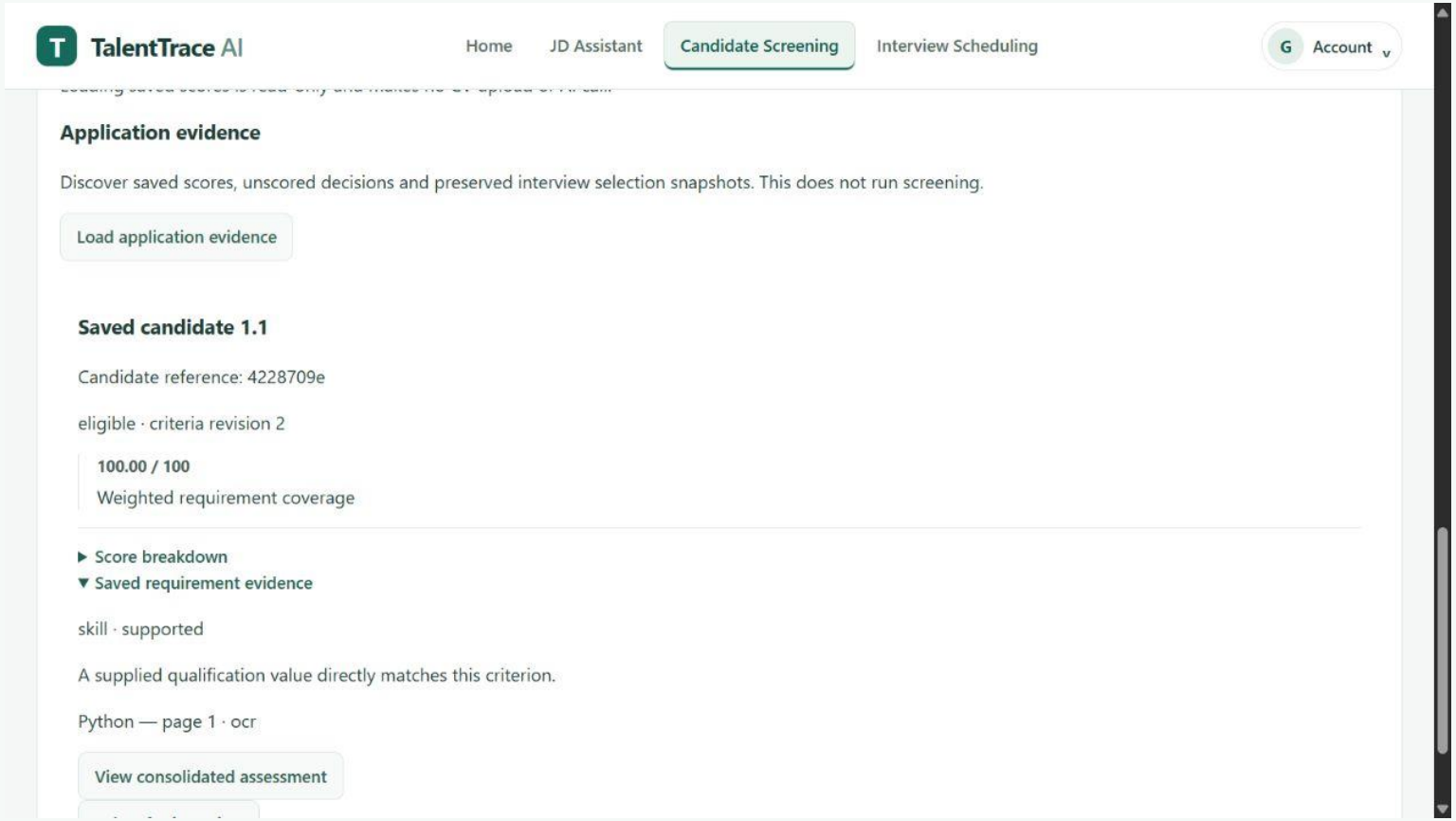
Job Reference remains stable.

Revision changes trigger renewed criteria review.

Historical capture before the final chat-layout polish

UI demonstration: screening evidence

Saved synthetic OCR result in the actual recruiter application. This is one example, not full-batch coverage.



Evidence before selection

Saved criteria revision

Requirement coverage

Source page and OCR provenance

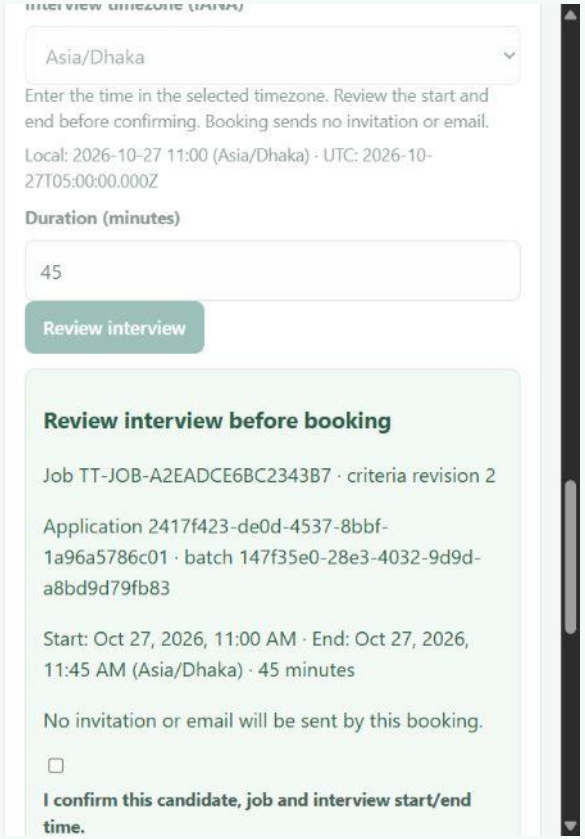
Recruiter review remains required

An unscored result is not a zero score

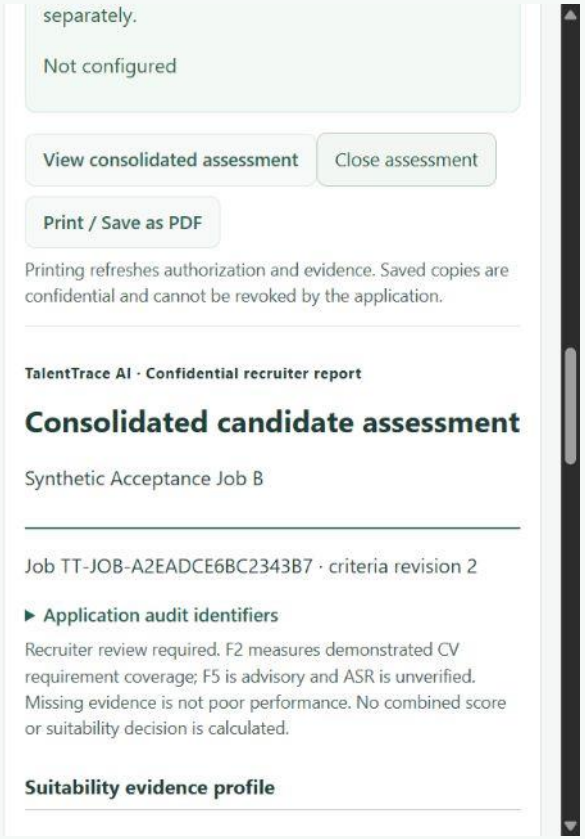
UI demonstration: booking and assessment

Actual synthetic application captures. Booking review and consolidated evidence are separate decisions.

Review before booking



Assessment and print



What these screens show

Review date, start/end, timezone and duration.

Booking sends no automatic email.

The report separates CV evidence from available interview evidence.

No invented combined hiring score.

Historical synthetic fixtures. No new booking or email.

Challenges and solutions

Recorded engineering problems, targeted fixes and limitations.

JD refinement ambiguity	Grounding/negation rule could reject valid changes.	Targeted validation fix; real Groq revision 1→2.	Resolved in synthetic JD test
Scanned / mixed PDFs	Image-only pages are not ordinary PDF text.	Tesseract OCR integrated into evidence-backed screening.	Verified on synthetic PDFs
AI provider configuration	Provider-disabled demo hid implemented Groq paths.	Separate enabled runtime; explicit cloud-data consent.	Groq synthetic flow verified
Interview integration	LiveKit/observer, consent and startup boundaries were fragile.	Working Cloud F3; real F4 audio reached provider but produced no recognized captions.	Media verified; combined STT chain partial
External integrations / privacy	Imports, posts and biometric experiments need distinct evidence.	One approved Google import and Discord post; face identity unverified.	Scoped imports/posts only

Current status and future improvements

Evidence through 28 October 2026. Separate successful trials do not prove the full live chain.

CURRENT VERIFIED EVIDENCE

- Real Groq JD refinement and saved revision
- One approved Google CV import; Discord post
- F3 two-person Cloud media (generated A/V)
- F6 evidence report and browser PDF export

Bounded Evidence

- N1: 20 parsed, 16 scored
- F4: live browser produced and recognized captions
- F5: model double; F11: STT browser acceptance
- N19 broad deletion and N21 second-machine proof remain partial

NEXT IMPROVEMENTS

- Complete candidate-associated erasure
- Verify a complete 20-CV provider run
- Complete real browser F4-to-F5/F11 acceptance
- Verify a second clean machine and approved remote HTTPS setup

LangGraph beyond a single LLM call

Two separate graphs serve different decisions, each with explicit validation and abstention.

F5 • POST-INTERVIEW EVALUATION

Pair Q&A
Speaker-attributed, finalized transcript

Map requirements
Confirmed job criteria revision

Retrieve reference
Curated local Python documentation

Assess + validate
Grounding checks; bounded abstention

F11 • LIVE RECRUITER CO-PILOT

Validate context
Run identity, consent and permissions

Assemble dialogue
Trusted role/time caption events

Track exploration
Answered vs uncovered requirements

Select + validate
Provisional follow-up or abstain

Privacy and human oversight

Recruitment evidence is sensitive; absence of evidence must never become a negative hiring score.

Authorization and scope

Recruiter/job/interview ownership checks; candidate one-use invitation; no cross-owner evaluation.

Separate consent

External Groq, caption processing, transcript storage and F11 assistant have independent gates.

Responsible scoring

Job-relevant evidence and source-linked requirements. Unscored is not zero; missing F5 is not failure.

Deletion boundary

Transcript withdrawal is implemented, but broad candidate-associated N19 deletion remains partial.

Use synthetic examples in demonstrations. Keep private CVs, transcripts, tokens and recordings outside Git.

Docker and reviewer setup

A clean checkout can run the provider-disabled application after explicit setup; full-feature parity is not verified.

Reviewer checkout (access to private repository required)

```
git clone --branch v1.0-final --single-branch https://github.com/shafi-mahmud/TalentTrace-AI.git
cd TalentTrace-AI
```

Baseline setup

New private config, fresh volume, migration 0017, Compose startup and synthetic recruiter provisioning.

Optional integrations

Groq needs private key + BGE model cache. LiveKit, AssemblyAI, F5 and F11 need separate readiness/consent.

Q&A

Thank you.

TalentTrace AI

1. Why LangGraph rather than one LLM prompt?
2. How do scores remain evidence-backed?
3. How is candidate privacy protected?
4. Which features are still not live-verified?

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